

Purr-fectly Average: Teacher Key & Notes

Answers, reasoning, and how to run the lesson · *About Average* (NRICH) extension



At a glance

- **The big idea (NRICH extension):** students explore *when* a set of conditions can be met and when it cannot, and build impossible conditions that look possible.
- **Time:** about 50–60 minutes for Parts 1–4; the Bonus fits as an extension or follow-up.
- **Interactive version:** milligan-mathematics.github.io/courses/253/about-average.html (solve cases with live checking, test any case, see a map of every case, design a case).
- **Original problem and teachers' resources:** nrich.maths.org/problems/about-average.
- **Cat theme:** numbers are mice caught by each cat, so "mean = fair share", "median = middle cat", "range = best hunter minus worst" all have a physical picture.



The mode rule (say this out loud)

The handout uses **one mode**: the single most-common count, shared by at least two cats. That is the rule the NRICH solution counts follow (on the interactive page Case 7 has two answers and Case 8 has three; the handout's warm-up Case 3 is that Case 8). It matters in Part 2 Case 4 and Part 3: with *tied* modes allowed, four cats with mean 5 and mode 9 becomes possible (1, 1, 9, 9). Ask students whether that set *should* count. There is no single right answer, but they must agree on the definition before arguing about the case.



Part 1 · Warm-up

Case	Answer(s)	Notes
1	2, 2, 5	Only set. Total = $3 \times 3 = 9$; two cats share 2, so the third has 5.
2	6, 6, 8, 10	Only set. Total 30. Mode 6 needs two 6s; the middle two must add to 14, so 6 and 8; the last is 10.
3	1, 7, 7, 9, 11 2, 7, 7, 7, 12 3, 5, 7, 7, 13	Three sets. Total 35. Students often find one and stop: ask "How do you know there aren't more?"

Expected strategy: turn the mean into a total first (mean $\times$ cats). Watch for students who average the clues instead of using them.



Part 2 - Predict, then test

#	Case	Verdict	Set and/or reason
1	Three cats. Mean 4, mode 2.	Possible	2, 2, 8 (the only set). Total 12; two cats share 2.
2	Three cats. Mean 4, range 12.	Impossible	Total 12. A range of 12 means the top cat has at least 13 (the lowest has at least 1), already more than 12 mice.
3	Three cats. Mean 7, mode 10.	Possible	1, 10, 10 (only set). A mode bigger than the mean is fine: one very low cat balances two high ones.
4	Four cats. Mean 5, mode 9.	Impossible (one-mode rule)	Total 20. Two 9s use 18, so the other two cats have 2 mice between them: 1 and 1. That makes 1 a second mode, so there is no single mode. Possible if ties are allowed: 1, 1, 9, 9.
5	Three cats. Mean 5, median 7, range 2.	Impossible	Sneaky: mean and median alone give exactly one set, 1, 7, 7 (total 15, middle 7, top at least 7, so the lowest is 1 and the top is 7). Its range is 6, not 2. Each clue is fine on its own; the three together clash.
6	Five cats. Mean 4, median 6.	Possible	1, 1, 6, 6, 6 (only set). Total 20. Three cats are at least 6 (the middle and everyone above), so they use at least 18, leaving exactly 1 and 1.
7	Five cats. Mean 4, median 6, range 6.	Impossible	Case 6 shows mean and median force 1, 1, 6, 6, 6, whose range is 5, not 6.
8	Three cats. Mean 4.5, mode 4.	Impossible	Total would be $3 \times 4.5 = 13.5$, but whole numbers of mice always add to a whole number. The mode clue is a distraction.

The surprises: Cases 3 and 6 are possible even though they look odd (a mode bigger than the mean; a median bigger than the mean). Cases 5 and 7 are impossible even though every clue alone is fine. Students who decide “impossible” only because they failed to find a set should be pushed for a reason: that is the heart of this lesson. Case 7 is a good one to revisit right after Case 6.



Part 3 - Move one dial

Mode	Three cats, mean 3 (total 9)	Four cats, mean 3 (total 12)
1	1, 1, 7	1, 1, 1, 9 (or 1, 1, 2, 8; 1, 1, 3, 7; 1, 1, 4, 6)
2	2, 2, 5	2, 2, 2, 6 (or 1, 2, 2, 7; 2, 2, 3, 5)
3	3, 3, 3	3, 3, 3, 3 (or 1, 3, 3, 5; 2, 3, 3, 4)
4	1, 4, 4	1, 3, 4, 4
5	X impossible	X impossible (1, 1, 5, 5 has two modes)
6	X impossible	X impossible (6, 6 already uses 12, leaving nothing)

- **a.** The biggest is 4 (1, 4, 4). A mode of 5 needs two 5s, which is 10 mice, more than the total of 9. (Even a mode of 4 leaves just 1 mouse for the third cat: 1, 4, 4.)
- **b.** Mean 4: **5** (2, 5, 5). Mean 5: **7** (1, 7, 7). Mean 6: **8** (2, 8, 8).
- **c.** The biggest mode is $(3M - 1) \div 2$, rounded down. (Checked for $M = 1$ to 10.) Reasoning: two cats share the mode m and the third needs at least 1, so $2m + 1 \leq 3M$. Accept any correct description in words.
- **d.** Two modes (1 and 5) tie, so neither is *the* mode under the one-mode rule. With ties allowed, 1, 1, 5, 5 works and the four cats with mean 3 can have a mode of 5 (but 6 is still out).



Part 4 - Design your own

- **Step 1** is automatically possible: the clues come from the secret set. The interesting question is *how many* sets fit. Fewer clues usually means more sets. Ask: "Which clue could you remove without changing the answers?" (The site's *Design a case* tab answers this.)
- **Step 2** is the hard part. A good impossible case passes the "looks possible" test: every single clue is reasonable, and ideally every pair is too. Watch for students who write something silly (a mean of 4.5 with three cats): push them to hide the reason. Good examples: Part 2 cases 5 and 7; or four cats, mean 6, median 5, range 3 (every pair of those clues works, but all three together do not).
- **Proofs:** accept any valid argument. The toolbox covers the usual reasons. The strongest students will argue from the total, the smallest possible values of the other cats, and the definition of the median.
- **Step 3** turns each student into a judge. Disagreements are productive. Usually one student has an impossible case that is in fact possible, or a "possible" case that is not.



Bonus - Bigger litters

Cats	Answer	Reason
3	Impossible	Two separate clashes. Median 2 and mode 1: the mode needs two 1s, which would push the middle cat down to 1. Mean 4 and range 10: total 12, and a range of 10 means 1 and 11 at least, leaving 0 for the third cat.
4	1, 1, 3, 11	Only set. Range 10 with mode 1 means lowest 1, top 11; total 16 leaves 4 for the middle two, which must be 1 and 3.
5	1, 1, 2, 5, 11	Only set.
6	1, 1, 1, 3, 7, 11	Only set. (Seven cats have five answers, eight cats have ten: the more cats, the more room.)

100 cats is possible. Example: fifty cats with 1 mouse, one cat with 3, forty-eight cats with 7, and one cat with 11. Check: total $50 + 3 + 336 + 11 = 400$ (mean 4); the 50th and 51st cats have 1 and 3 (median 2); 1 appears 50 times, more than anything else (mode 1); range $11 - 1 = 10$. The site's *stacking view* lets students build it in a minute.



Questions to keep asking

- What makes each "average" different from the others?
- Are there other sets? How do you know you have them all?
- Which clue did you use first? Which clue is most likely to make a case impossible?

- Can you give a reason, not just “I couldn’t find one”?
- Is this impossible because of one clue, or because of a combination?

Support: give partial sets to complete (for example “2, __, 5” and a mean) before asking students to build from nothing, as in the NRICH teachers’ notes. *Extend:* Part 4 and the Bonus; or have students explore the Possibility Map and describe the “one-answer” diagonal.

Cases 1–8 of the interactive page are the NRICH puzzles: nrich.maths.org/problems/about-average. All answers above were checked by exhaustive search.